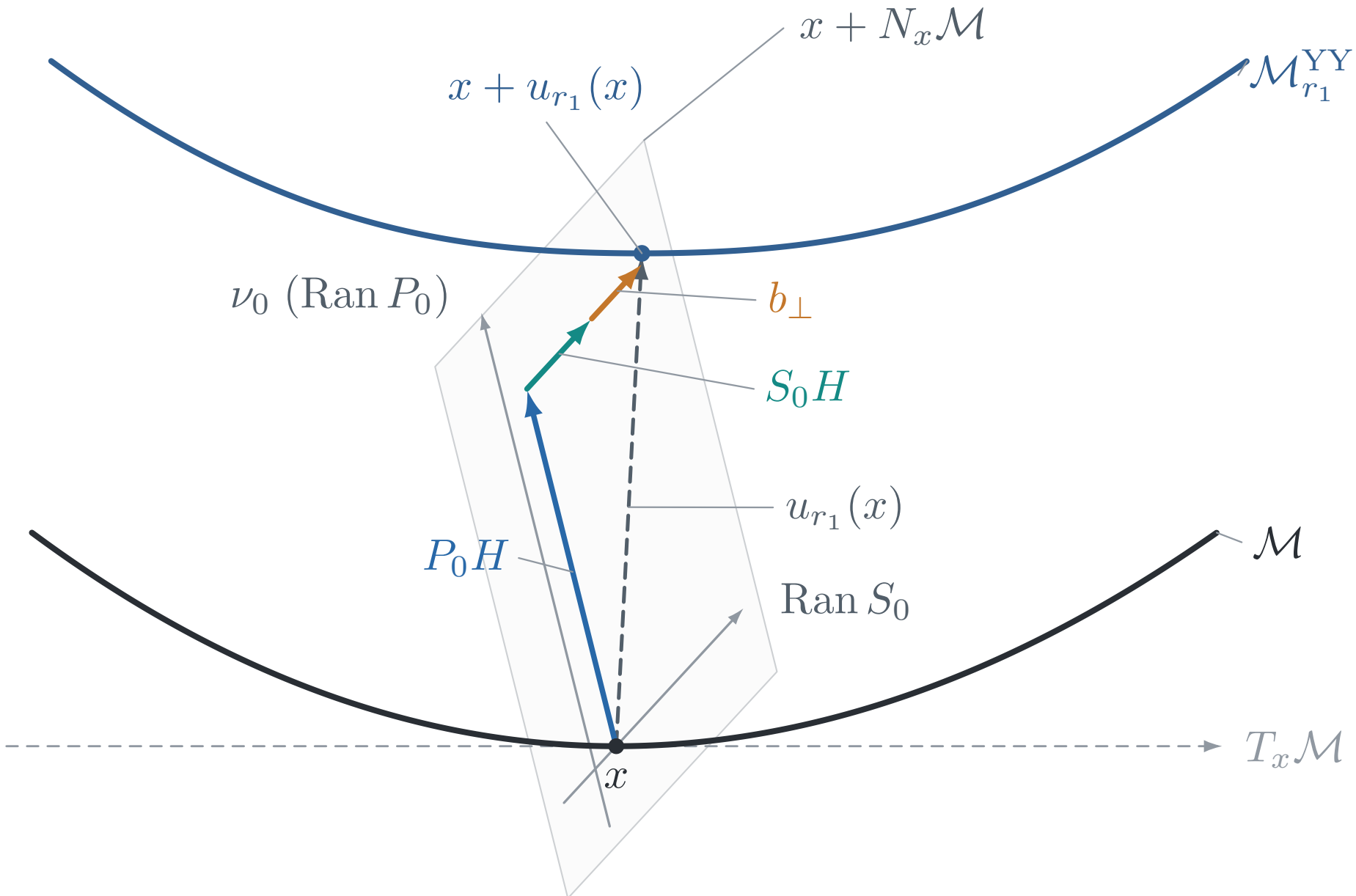

$x + N_x\mathcal{M}$
$x + u_{r_1}(x)$
$\mathcal{M}_{r_1}^{\mathrm{YY}}$
$\nu_0\ (\mathrm{Ran}\, P_0)$
$b_\perp$
$S_0 H$
$u_{r_1}(x)$
$P_0 H$
$\mathcal{M}$
$\mathrm{Ran}\, S_0$
$T_x\mathcal{M}$
x